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September 1, 1975
Superseding
MIL-STD-1855A
May 9, 1975

FEDERAL STANDARD

TAPE, AUDIO, MAGNETIC, 1/4-INCH, RECORDING FORMATS FOR

Authority. This standard is issued pursuant to the Federal Property and Administrative Services Act of 1949, as amended, and its application to both magnetic audio tape recording and magnetic audio tape recording/reproducing equipment is mandatory on all Federal Agencies.

S1. Purpose. This standard establishes standard recording formats for use by all Federal Agencies in the recording of 1/4-inch (6.3mm) magnetic audio tape.

S1.1 Scope. This standard provides media physical characteristics and detail recording formats for 1/4-inch (6.3mm) magnetic audio tape recording applications.

S1.2 Application. Procurement of 1/4-inch (6.3mm) magnetic audio tape recording and reproducing equipment shall be limited to equipment that records and reproduces in accordance with the recording formats and other characteristics given herein. Existing Government stocks of equipment using other recording formats may continue to be utilized until exhausted.

S1.3 Format identification. The standard audio recording formats shall be identified by type, as follows:

Type I	-	Reel-to-Reel Full Track Monophonic
Type II	-	Reel-to-Reel Two Track Monophonic
Type III	-	Reel-to-Reel Two Track Stereophonic
Type IV	-	Reel-to-Reel Four Track Monophonic
Type V	-	Reel-to-Reel Four Track Stereophonic
Type VI	-	Endless Loop Cartridge Two Track Monophonic
Type VII	-	Endless Loop Cartridge Three Track Stereophonic

S1.3.1 Class. Type VI and Type VII cartridges shall be identified by class, based on cartridge size (see table I of figures 1 thru 4) and playing time at 7.5 in/s (19.05 cm/s), as follows:

Class	Playing time @ 7.5 in/s (19.05 cm/s)
A	15 minutes
B	30 minutes
C	60 minutes

S2. Referenced Documents.

S2.1 The following documents of the issue in effect on the date of invitation for bids form a part of this standard to the extent specified herein.

Federal Specifications:

W-R-175	-	Reels and Hubs for Magnetic Recording Tape (General Specification for).
W-R-175/1	-	Reels, Plastic and Fiberglass, With 5/16-Inch Center Hole (Detail Specification).
W-R-175/3	-	Reels, Plastic, Fiberglass, Metallic, and Metallic Flange Phenolic Hub, With 3-Inch Center Hole (Detail Specification).
W-R-001572	-	Tape, Recording, Audio, 1/4-Inch, Magnetic Oxide-Coated.

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Services Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

S2.2 Other publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply.

National Association of Broadcasters (NAB) Standards:

NAB Standard - Cartridge Tape Recording and Reproducing, October 1964.

(Application for copies should be addressed to the National Association of Broadcasters, 1771 N Street, NW, Washington, DC 20001.)

S3. Media Physical Characteristics and Recording Formats.

S3.1 General Requirements.

S3.1.1 Tape width. Tape width for both reel-to-reel recording and endless-loop-cartridge recording shall be 0.246 inch ± 0.002 inch (6.25mm ± 0.05mm).

S3.2 Reel-to-Reel Physical Characteristics.

S3.2.1 Tape lengths. Standard tape lengths are in accordance with W-T-001572 as follow.

Reel diameter (inches)	Hub Center hole (inches)	1.5 mil base minimum length (feet)	1.0 mil base minimum length (feet)
3 (76.2mm)	5/16 (8.1mm)	155 (47.24m)	----------
4 (101.6mm)	5/16 (8.1mm)	305 (92.96m)	----------
5 (127.0mm)	5/16 (8.1mm)	610 (185.93m)	915 (278.90m)
7 (177.8mm)	5/16 (8.1mm)	1215 (370.33m)	1820 (554.74m)
10.5 (266.7mm)	3 (76.2mm)	2520 (768.10m)	3620 (1103.38m)
14 (355.6mm)	3 (76.2mm)	5025 (1531.60m)	7230 (2203.70m)

S3.2.2 Tape reel dimensions. Tape reel dimensions shall be in accordance with W-R-175/1 and /3 as applicable.

S3.2.3 Tape wind. Tape shall be wound with the oxide coated surface facing toward the hub of the reel. Recorded tape should be wound so that the start of the program material is at the outside of the reel.

S3.3 Reel-to-reel tape speeds. The preferred tape speed shall be 7.5 in/s (19.05cm/s) ± 0.2 percent. Supplementary tape speeds shall be 15 in/s (38.1 cm/s), 3.75 in/s (9.53 cm/s), and 1.875 in/s (4.76 cm/s) ± 0.2 percent. The tolerance on tape speed shall apply to any portion of the reel of tape in use.

S3.4 Position and Dimensions of Magnetic Tracks on Reel-to-Reel Tape.

S3.4.1 Magnetic track designations. Track 1 shall be the top track when the tape is moving from left to right with the coated side facing away from the observer and with the leader to the right. The next lower tract is designated Track 2, and so on.

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S3.4.2 Magnetic track dimensions. The recorded track width for full track recordings shall be 0.238 inches + 0.010, -0.004 inches (6.05mm + 0.25, -0.10mm). The recorded tracks for two track recordings shall be 0.082 inches ± 0.002 inches (2.08mm ± 0.05mm) in width with a center-to-center spacing of 0.156 inches ± 0.004 inches (3.96mm ± ± 0.10mm). The recorded tracks for four track recordings shall be 0.043 inches + 0.000, - 0.004 inches (1.09mm + 0.000, - 0.10mm) in width. The center-to-center spacing between Tracks 1 and 3, and between Tracks 2 and 4, shall be 0.134 inches + 0.002, - 0.000 inches (3.40mm + 0.05, - 0.00mm). Tracks 1 and 4 shall be so positioned that their outer edges coincide with the edges of the tape when the tape width is minimum.

S3.4.3 Reel-to-reel full track monophonic recordings - Type I format. A single track shall be recorded over the working width of the tape.

S3.4.4 Reel-to-reel two track monophonic recordings - Type II format. Two tracks shall be recorded in sequence in alternate directions, Track 1 first, Track 2 second.

S3.4.5 Reel-to-reel two track stereophonic recordings - Type III format. Two tracks shall be recorded in the same direction. Track 1 shall carry the recording for the left-hand channel as viewed from the listening position. Track 2 shall carry the recording for the right-hand channel. The tracks shall be recorded with head-gaps in line and phased for reproduction on equipment so connected that when a full track tape is reproduced it produces in-phase signals in the two channel outputs.

S3.4.6 Reel-to-reel four track monophonic recordings - Type IV format. The tracks shall be recorded in alternate directions in the sequence 1-4-3-2.

S3.4.7 Reel-to-reel track stereophonic recordings - Type V format. Tracks 1 and 3 shall be used simultaneously for one direction of tape travel and Tracks 2 and 4 for the other direction. Tracks 1 and 4 shall carry the recording for the left-hand channel, as viewed from the listening position, Tracks 2 and 3 shall carry the recording for the right-hand channel. The tracks shall be recorded with the head-gaps in line and shall be phased for reproduction on equipment so connected that when a full track tape is reproduced it produces in-phase signals in the two channel outputs.

S3.5 Cartridge Physical Characteristics.

S3.5.1 Tape lengths. Tape lengths shall be in accordance with S1.3.1 and shall conform to the cartridge loading requirement of the NAB Standard - Cartridge Tape Recording and Reproducing, October 1964.

S3.5.2 Cartridge dimensions. Cartridge dimensions shall be in accordance with figures 1 thru 4 and table I therein. The dimensional drawings in figures 1 thru 4 are in accordance with cartridge dimensional drawings in the NAB Standard - Cartridge Tape Recording and Reproducing, October 1964.

S3.5.3 Tape wind. The tape shall be wound with the oxide coated surface facing away from the hub of the tape turntable contained within the cartridge.

S3.6 Cartridge tape speed. The tape speed shall be 7.5 in/s (19.05 cm/s) with a speed accuracy of ± 0.4 percent as measured over a 150 foot (45.7m) ± 1.0 inch (25.4mm) loop of one mil (base film thickness) lubricated tape loaded in a class A cartridge.

S3.7 Position and Dimensions of Magnetic Tracks on Cartridge Tape.

S3.7.1 Cartridge two track monophonic recordings - Type VI format. Track positions and dimensions shall be in accordance with figure 5. The drawing in figure 5 is in accordance with Chart A in the NAB Standard - Cartridge Tape Recording and Reproducing, October 1964.

S3.7.2 Cartridge three track stereophonic recordings - Type VII format. Track positions and dimensions shall be in accordance with figure 6. The drawing in figure 6 is in accordance with Chart C in the NAB Standard - Cartridge Tape Recording and Reproducing, October 1964.

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S4 Changes. When a Federal agency considers that this standard does not provide for its essential needs, written request for changing or adding to the standard, supported by adequate justification explaining wherein the standard does not provide for essential needs, shall be sent in duplicate to the General Services Administration, Federal Supply Service, Office of Standards and Quality Control, Washington, DC 20406. The General Services Administration will determine the appropriate action to be taken and will notify the agency.

S5 Conflict with referenced documents. Where the requirements stated in this standard conflict with any requirement in a referenced document, the requirements of this standard shall apply. Nature of conflict between the standard and the referenced document shall be submitted in duplicate to the General Services Administration, Federal Supply Service, Office of Standards and Quality Control, Washington, DC 20406.

S6 Supersession data. This standard supersedes military standard MIL-STD-1855A, 9 May 1975.

Military Custodians:

Army - EL
Navy - EC
Air Force - 80

Review Activities:

Navy - AS
Air Force - 84

User Activities:

Army - MI, AV
Navy - TD

Preparing Activity:

GSA-FSS

Civil Agency Coordinating Activities:

Commerce
CSC
DOT
EPA
Energy Research Dev. Adm.
FCC
GSA-FSS
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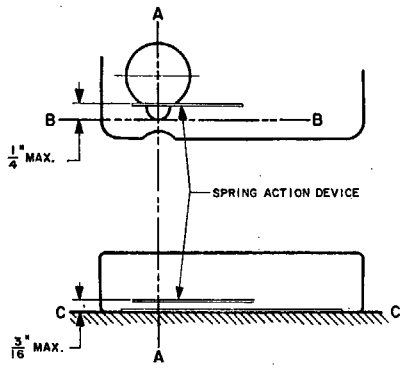


FIG. 1 REQUIRED SPRING ACTION DEVICE LIMITATIONS

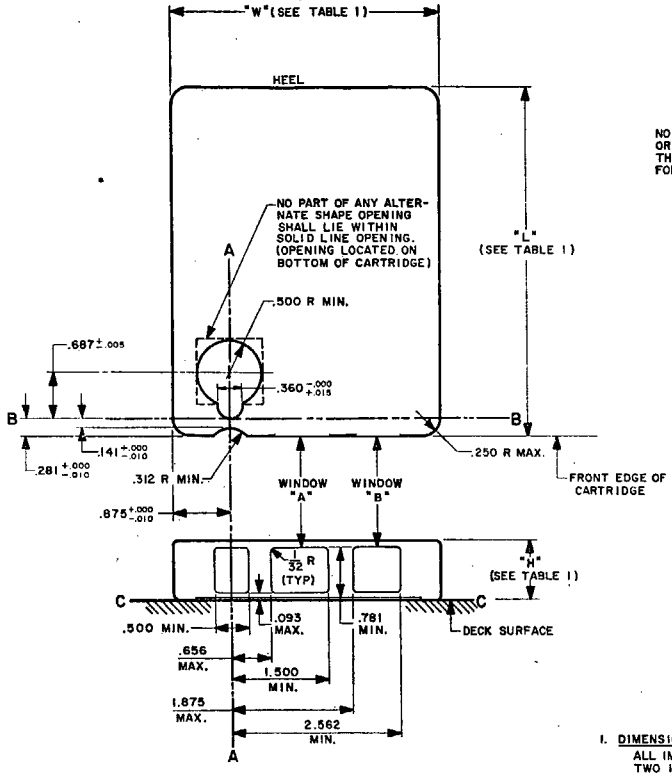


FIG. 3 CARTRIDGE DIMENSIONS

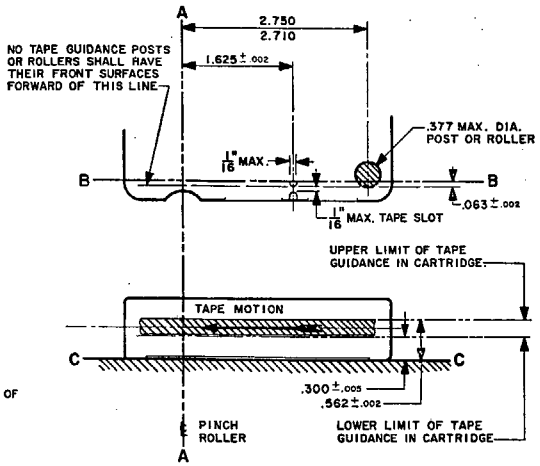


FIG. 4 TAPE GUIDANCE LIMITS

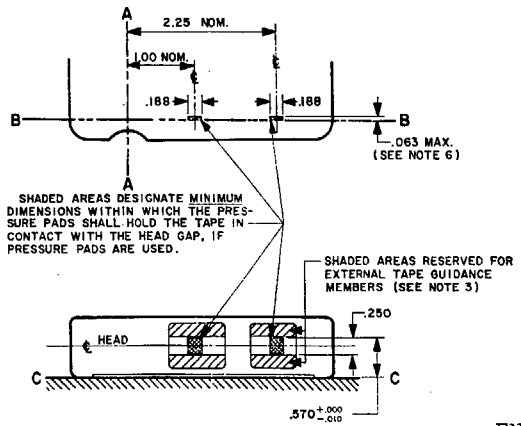


FIG. 2 PRESSURE PAD PROVISION (IF USED)

TABLE I

CARTRIDGE NAB TYPE	WIDTH "W" ± 1/64"	LENGTH "L" MAX.	HEIGHT "H" MAX.
A	4"	5 1/4"	.9375
B	6"	7"	.9375
C	7 5/8"	8 1/2"	.9375

FIGURES 1 THRU 4
ENDLESS LOOP TAPE CARTRIDGE DIMENSIONS

NOTES

1. DIMENSIONS:
   ALL IMPORTANT OPERATING DIMENSIONS ARE REFERENCED FROM TWO IMAGINARY MUTUALLY PERPENDICULAR, VERTICAL PLANES MARKED A-A & B-B, AND A THIRD HORIZONTAL PLANE C-C, REPRESENTING THE DECK SURFACE OF THE CARTRIDGE TAPE PLAYER.

   WHERE APPLICABLE, DIMENSIONS INCLUDE DRAFT ALLOWANCES.

2. MATERIALS:
   ALL MATERIALS USED IN THE CARTRIDGE CONSTRUCTION SHALL BE NON-MAGNETIC.

3. TAPE GUIDANCE:
   PRECISE TAPE GUIDANCE SHALL BE PROVIDED BY TAPE GUIDE, OR GUIDES, EXTERNAL TO THE CARTRIDGE. INTERNAL GUIDANCE, AS SPECIFIED, IS TOLERANCE AND DIMENSIONED SO AS NOT TO INTERFERE WITH EXTERNAL HEAD GUIDES, PROPERLY SET, AND USING ST'D 1/4" TAPE (.246±.002)

   ROLLERS, POSTS, OR OTHER TAPE GUIDING SURFACES MUST BE SMOOTH AND PROVIDE A VERTICAL TAPE PLANE IN THAT PORTION OF TAPE WHICH PASSES ACROSS THE FRONT OF THE CARTRIDGE.

   THE MINIMUM WINDOW AREA, EXCEPT FOR THE AREA COVERED BY THE TAPE PATH, SHALL BE KEPT CLEAR FOR A DISTANCE OF .187 TO THE REAR OF B-B TO PROVIDE FOR EXTERNAL TAPE GUIDES.

4. TAPE TENSION:
   TAPE TENSION SHALL NOT EXCEED 8 OZ. AT CAPSTAN.

5. TAPE LENGTH:
   TAPE LENGTH IN PLAYING TIME SHALL BE LEGIBLY MARKED ON THE HEEL OF CARTRIDGE BASE.

6. HEAD PENETRATION:
   HEAD PENETRATION SHALL CONFORM TO CROSS-HATCHED AREA SHOWN IN FIG. 2.

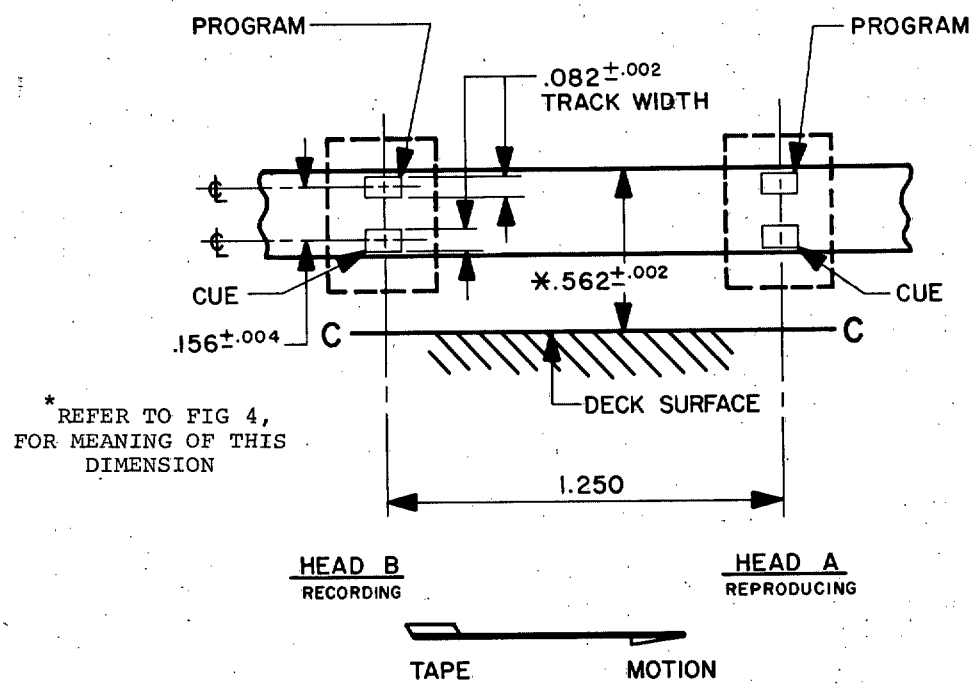
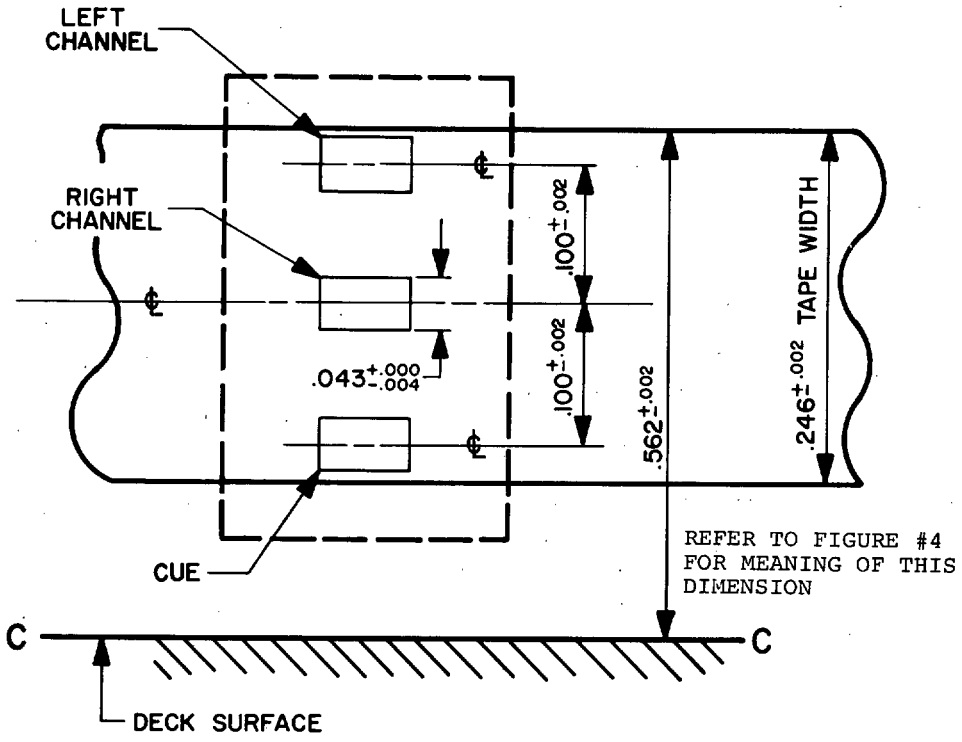


FIGURE 5
ENDLESS LOOP TAPE CARTRIDGE
MONOPHONIC TWO-TRACK RECORDED TRACK DIMENSIONS

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NOTES:

1. IF SECOND STEREO HEAD IS USED IT SHOULD BE PLACED 1.250 FROM FIRST HEAD.

2. TRACK WIDTH: ALL TRACKS SHALL BE $.043^{+.000}_{-.004}$

FIGURE 6
ENDLESS LOOP TAPE CARTRIDGE STEREO 3-TRACK
RECORDED TAPE TRACK DIMENSIONS

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